Cynthia Chen

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Education

Rutgers University - New Brunswick - *B.S. Computer Science, B.A. Cognitive Science (Minds, Machines and Computation track)*

SEP 2020 - MAY 2024

- GPA: 3.95
- Dean's List, Graduated with Honors

Research Experience

Allen Institute for Data Driven Discovery in Biology
Postbac Research Fellow - Cindy Poo and Michael Buice

JUN 2024 - JUN 2026
Seattle, WA

- Designed a neural network architecture that incorporated classical computation algorithms to solve problems of decision making under uncertainty, outperforming other models in the same class
- Presented the emergence of interpretable value representations in novel machine learning architecture at ICLR conference
- Designed Markov models to represent state-mediated decision making and belief update processes observed in rodent experiments
- Used fitting techniques such as regression and likelihood estimation to understand model's representations of experimental data

NeuroCognition Lab
REU Research Assistant - Adham Atyabi

JUN 2023 - NOV 2023
Colorado Springs, CO

- Applied signal processing and dimensionality reduction techniques on EEG and ECoG data such as Fourier transforms and Principle Component Analysis with EEGLab and mne
- Designed Convolutional neural networks and transformer models for EEG decoding with Brain Computer Interface applications that competed with the state of the art models in classification accuracy
- Presented research and findings to REU cohort students and authored a symposium paper detailing experimental methodology and results.

Social Entrepreneurship, Networks, Digital Innovation Lab
Research Assistant - Sophia Fu

JUN 2021 - MAY 2023
New Brunswick, NJ

- Awarded Aresty Summer Science grant to work on research project and presented poster at corresponding Symposium
- Conducted exploratory analysis and literature review for abstract presented at International Communication Association Conference
- Analyzed and performed manual coding of over 50 papers and interview transcripts
- Recruited hundreds of organizations for research studies on nonprofit resilience and stakeholder management

Skills

Coding - python, java, matlab-EEGLab, git
Machine Learning - Docker, cloud computing (CodeOcean), Pytorch-CUDA
Scientific Communication - Paper writing, Oral Presentations, Poster design and presention

Publications and Presentations

Chen, C., Johnson, S., Poo, C., Buice, M., & Khalvati, K. (2026). "Information-based Value Iteration Networks for Decision Making Under Uncertainty" *International Conference on Learning Representations 2026*

Chen, C., Buice, M., & Khalvati, K. (2025). "Factorized Value Iteration Network for Decision Making under Mixed Observability." *Generalization in Planning Workshop, AAAI 2025*.

Lin, H., Zhang, R., Scrivani, K., **Chen, C.**, Fu, S.,(2024). "Mapping the Landscape of Online Discourse on Climate Change-displaced Populations: A Topic Modeling Analysis with BERTopic." *74th Annual ICA Conference*

Chen, C., & Atyabi, A. (2023)." Finger Flex Classification for Brain Computer Interfaces." *REU Symposium on Deep Learning.*

Relevant Community Service

BrainFest - Pacific Science Center NOV 2024
Science Booth Coordinator
- Facilitated interactive booth teaching visitors mainly aged 5-15 about EMG signals and technologies that utilize them

Rutgers Cognitive Science Club SEP 2021 - MAY 2024
Executive Board, Graphic Design
- Designed graphics to promote weekly events
- Assisted marketing team to ensure events are properly represented in promotional material
- Collaborated with executive board members and external organizations to plan and organize club events
- Researched and create interactive presentations on topics in cognitive science to general club members

Rutgers Undergraduate Research Journal SEP 2021 - MAY 2022
Peer reviewer
- Contributed to intensive peer review process of three papers for an interdisciplinary undergraduate journal
- Collaborated with editorial committee on curation of accepted papers for the journal issues